

Aditya Roy

+91-9142560256 | adityaroyofficial20@gmail.com | [LinkedIn](#) | [GitHub](#)
[LeetCode](#) | [GeeksForGeeks](#) | Bhubaneswar, India

EDUCATION

- **Kalinga Institute of Industrial Technology** 2023 – Present
Bachelor of Technology in Computer Science CGPA: 9.33
- **U.R. College, Rosera** 2020 – 2022
Higher Secondary School Percentage: 80.40%
- **Sun Shine School, Rosera** 2018 – 2020
Senior Secondary School Percentage: 92.20%

EXPERIENCE

- **Wushu Sports Management Platform** [\[Live\]](#) Aug 2025 – Present
Lead Full-Stack Developer (Node.js, MongoDB, React) Team of 5
Node.js, Express.js, MongoDB, React, Zustand, JWT Authentication, RBAC, SSE, Winston Logging, Cloudinary
 - **Backend Architecture:** Built a 233-endpoint REST API powering a **multi-role** championship management system, structured around **event-scoped** workflows for registrations, approvals, fixtures and grievances, across national, state, and district levels.
 - **Authorization System:** Engineered a database-driven **RBAC system** with hierarchical roles and **jurisdiction-based** filtering, enforcing fine-grained access control across event-specific operations.
 - **Database Design:** Designed a MongoDB schema with 30+ collections, using aggregation pipelines, indexing, pagination, and selective denormalization to efficiently handle large-scale athlete and registration data
 - **Workflow & Validation:** Implemented **multi-stage submission and approval** pipelines (team manager → coordinator → admin) with athlete-level verification, driven by a configurable circular system defining categories, rules, and fees.
 - **Security & Reliability:** Developed JWT-based authentication with httpOnly cookies and refresh-token rotation, layered validation, centralized error handling, and structured logging using Winston.
 - **Real-Time Features:** Built real-time notification pipelines using Server-Sent Events (SSE) and email workflows for live updates across registrations and championship operations.
 - **Deployment:** Deployed the full-stack application on Render with Cloudinary-based media storage and environment-based production configuration.

PROJECTS

- **Transformer Intelligence Desk (Agentic QA)** [\[GitHub\]](#) Apr 2026
LangGraph, Groq (Llama-3), ChromaDB, SentenceTransformers, Python, Streamlit
 - **Agentic RAG Architecture:** Architected an intelligent QA system using **LangGraph** and **Llama-3.3-70b** to dynamically route complex research queries between knowledge retrieval, mathematical calculation, and conversational memory.
 - **Vector Search & Tool Calling:** Built a specialized **ChromaDB** vector store with hybrid multi-query retrieval and integrated a secure Python sandbox to calculate exact hyperparameter math, effectively eliminating LLM hallucinations.
 - **Automated Self-Correction:** Engineered a continuous evaluation loop (**LLM-as-a-Judge**) that grades response faithfulness, triggering prompt adjustments and generation retries if the score falls below a strict 0.7 threshold.
 - **State Management & Resilience:** Implemented **MemorySaver** for persistent multi-turn context with explicit identity extraction, and designed a seamless API fallback pool to guarantee high availability during rate limits.

SKILLS

- **Languages:** C, C++, Java, Python, JavaScript, HTML, CSS
- **Backend & Frameworks:** Node.js, Express.js, REST API Design, LangGraph
- **Frontend:** React, Zustand, Redux Toolkit, React Hook Form, Tailwind CSS
- **Databases:** MongoDB, Mongoose, ChromaDB (Vector Search), NoSQL Data Modeling, Aggregation Pipelines
- **Tools & Concepts:** Agentic AI, RAG, JWT Auth, RBAC, Git, GitHub, Render, Cloudinary, Postman
- **Coursework:** Data Structures & Algorithms, Operating Systems, DBMS, Computer Networks

ACHIEVEMENTS

- **Amazon ML Summer School 2025** [\[Cred\]](#) Jul 2025 – Aug 2025
Selected Student Remote
 - **Selection:** Selected among top engineering students across India for Amazon's national ML mentorship program.
- **Competitive Programming** [\[Cred\]](#) Oct 2024
Top 10 Rank KIIT University
 - **Competition:** Ranked in the **top 10 out of 600+ participants** in a university-wide Data Structures and Algorithms competition.
 - **DSA:** Solved **700+ algorithmic problems** across **LeetCode (500+)** and **GeeksForGeeks (200+)**.